Relative Tree Gauge and Boundary-State Disintegration

Abstract

We give an exact finite-graph separation of an $SU(2)$ lattice connection into a boundary-covariant retained state and an interior conditional fiber. Interior gauge redundancy is removed by a boundary-rooted forest, while the boundary gauge action remains visible. This supplies the typed interface needed by state-weighted defect transfer.

1 Relative tree gauge

Let $\Gamma = (V, E)$ be a finite oriented graph with nonempty boundary ∂V. The relative gauge group is $G^{V\setminus\partial V}$, acting on $U \in G^E$ by

$$U_{xy} \longmapsto g_x U_{xy} g_y^{-1}, \qquad g_b = 1 \quad (b \in \partial V).$$

Choose a forest F containing one directed path from a boundary root to every interior vertex.

Theorem 1 (Relative tree-gauge coordinates)**.** *For every $U \in G^E$, there is a unique relative gauge transformation that sets every forest link to the identity. It induces a bijection*

$$G^E/G^{V\setminus\partial V} \cong G^{E\setminus F}.$$

The normalized nonforest variables are invariant under interior gauge transformations and covariant under independent transformations of their two boundary roots.

Proof. Set the normalizer at every boundary root to one and propagate it recursively along F: if v is the child of u, put $h_v = h_u U_{uv}$, reversing the link when required. Then $h_u U_{uv} h_v^{-1} = 1$. A forest has a unique root path, so the recursion and the normalizer are unique. Under an interior gauge g, the path product changes to $h_v g_v^{-1}$, which cancels in each normalized nonforest edge. If boundary gauges are allowed, the two root factors remain on the left and right. □

Proposition 1 (Haar disintegration)**.** *With normalized Haar measure,*

$$\prod_{e\in E} dU_e = \left(\prod_{v\in V\setminus\partial V} dg_v\right)\left(\prod_{e\in E\setminus F} dH_e\right).$$

Consequently any local plaquette density, including the heat-kernel density, has an exact conditional disintegration after the nonforest variables are partitioned into retained boundary variables σ and interior fiber variables ϕ.

Proof. The recursive change of variables uses only group multiplication and inversion. Haar invariance makes each triangular step unit-Jacobian. Fubini's theorem then gives the stated product disintegration. □

2 Sufficiency

For the reflection-symmetric 2×2 square, there are nine vertices, twelve links, four plaquettes, eight boundary links, one forest link, and three fiber links. A forest invariant under horizontal reflection makes relative normalization commute with that reflection.

The complete retained boundary connection determines every plaquette term crossing from the block to its exterior. Omitting a retained crossing edge is not sufficient: changing only that edge changes the holonomy, and hence any nonconstant central plaquette weight. Center sign, cut-locus proximity, and central versus regular stabilizer type are derived from retained Wilson loops; they are not discarded labels.

3 Scope

The finite representation theorem is exact. A complete 2^4 block, overlap-consistent coarse variables, a state weight, and a subcritical nonlinear bad kernel remain open. Reflection compatibility here does not by itself prove reflection positivity of an iterated RG map.